CRM Sales Pipeline Analytics – Project Documentation

End-to-end data analytics portfolio project | Domain: B2B Sales Operations / CRM

Tools: Excel, Power Query, PostgreSQL, Python, Power BI, DAX

1. Project Overview

This project analyzes a B2B customer relationship management (CRM) dataset of 8,800 sales opportunities across multiple products, customer accounts, sales agents, and regional offices.

The analysis covers revenue generation, sales conversion, pipeline health, sales team productivity, product performance, and CRM data quality. The deliverables are an interactive Power BI dashboard, supported by PostgreSQL analysis, Python exploratory data analysis, and validated DAX measures.

Headline results

Metric	Result
Total opportunities	8,800
Won revenue	$10,005,534
Closed-deal win rate	63.15%
Open opportunities	2,089
Open opportunities missing account info	1,425 (68.21%)

2. Business Problem and Objectives

Sales leadership needs reliable visibility into sales performance and the current opportunity pipeline. Without consolidated reporting, it is hard to know:

- Which products and regions generate the most revenue.
- Which sales agents and managers perform effectively.
- How successfully opportunities convert into won deals.

- Where open opportunities are concentrated.
- How long opportunities take to close.
- Whether incomplete CRM records affect customer-level reporting.

The goal is to turn transactional CRM data into a consistent reporting solution that supports performance monitoring and evidence-based decisions.

Objectives

1. **Revenue analysis:** total won revenue, average deal value, monthly trends, product contributions.
2. **Conversion analysis:** closed-deal win rates and won versus lost opportunities.
3. **Pipeline analysis:** open opportunities by deal stage, product, and regional office.
4. **Sales team analysis:** agent and manager comparison by revenue, conversion, and sales-cycle metrics.
5. **Customer analysis:** account-level and industry-sector revenue contributions.
6. **Data quality analysis:** missing customer information and its impact on reporting.

3. Stakeholders

These are representative stakeholders for this portfolio case study, not stakeholders from an actual commissioned project.

Stakeholder	Analytical requirements
Sales Director	Overall revenue, win rate, and regional performance
Regional Sales Manager	Regional pipeline, team performance, and opportunities
Sales Operations Analyst	Sales-cycle analysis, opportunity stages, and CRM data quality
Product Manager	Product revenue contribution and pipeline distribution
Account Manager	Customer-level performance and account visibility

4. Scope and Success Criteria

In scope

- Four related CRM source datasets.
- Data inspection and transformation using Power Query.
- Relational data modeling and DAX calculations.
- SQL-based business analysis in PostgreSQL.
- Python exploratory data analysis.
- Four interactive Power BI report pages.
- Business insights, recommendations, and metric validation.

Out of scope

- Predictive machine learning models.
- Automated CRM data ingestion.
- Real-time dashboard refresh.
- Production CRM integration.
- Automated sales forecasting.

Success criteria

- Core KPIs are defined consistently.
- PostgreSQL, Python, and Power BI produce matching validated results.
- Dashboard pages answer the defined business questions.
- Users can filter, explore, and drill through sales performance.
- Data quality risks are clearly identified.
- Findings are translated into actionable recommendations.
- The project is documented and organized for GitHub.

5. Technology Stack

Tool	Purpose
Excel	Source data files
Power Query	Data inspection and preparation for reporting

Tool	Purpose
PostgreSQL	Independent business analysis and metric validation
Python (Pandas, NumPy, Matplotlib, Seaborn)	Exploratory data analysis and verification
Jupyter Notebook in VS Code	Interactive analysis environment
Power BI and DAX	Data model, measures, and interactive dashboard

6. Repository Structure

The project is organized into the following folders and files:

Path	Contents
`Raw_Data/`	Original source datasets, preserved unchanged
`Power_Query/`	Excel files referenced by the Power BI report
`PostgreSQL/`	SQL scripts for analysis and validation
`Python/CRM_Sales_Pipeline_EDA.ipynb`	Exploratory data analysis notebook
`Python/requirements.txt`	Required Python libraries

7. Data Sources

7.1 Source Datasets

Dataset	Business purpose	Key columns
`sales_pipeline.xlsx`	Opportunities, deal stages, values, and dates	opportunity_id, sales_agent, product, account, deal_stage, engage_date, close_date, close_value

Dataset	Business purpose	Key columns
`accounts.xlsx`	Customer account and industry information	account, sector
`products.xlsx`	Product reference information	product
`sales_teams.xlsx`	Maps sales agents to managers and regional offices	sales_agent, manager, regional_office

`sales_pipeline` is the primary transactional table with 8,800 opportunities. The other three datasets provide descriptive attributes for analysis by customer, product, agent, manager, and region.

7.2 Data Granularity

The analytical grain is **one row per sales opportunity**, identified by `opportunity_id`. Each row holds the assigned sales agent, product, customer account, deal stage, relevant dates, and closing value. Maintaining this grain avoids duplicate opportunity counts and inflated revenue.

7.3 Opportunity Stages

Deal stage	Count	Classification
Won	4,238	Closed
Lost	2,473	Closed
Engaging	1,589	Open
Prospecting	500	Open
Total	**8,800**	

- Closed opportunities = Won + Lost = 6,711
- Open opportunities = Engaging + Prospecting = 2,089
- Won revenue = sum of closing values for Won opportunities
- Closed-deal win rate = Won / Closed (open opportunities are excluded because their outcomes are not yet determined)

8. Data Preparation

8.1 Excel and Power Query

Source files were inspected and prepared through Power Query. The workflow:

1. Review dataset structures and column data types.
2. Check missing values and opportunity classifications.
3. Ensure date and numeric fields are suitable for analysis.
4. Prepare datasets for relational modeling.
5. Validate business measures after loading.

8.2 Preparation Principles

- Preserve the original source files.
- Maintain one record per opportunity.
- Do not replace unknown account information with fabricated values.
- Keep open opportunities separate from completed sales.
- Use consistent KPI definitions across all tools.
- Validate totals before interpreting business performance.

8.3 Validation Rules Applied

- **Deal stage:** every record belongs to one of the four expected stages, and the open/closed distinction is applied consistently across SQL, Python, and Power BI.
- **Revenue:** only Won opportunities count as revenue. Open pipeline values are never mixed with completed sales, and lost deals contribute nothing.
- **Sales cycle:** duration runs from engagement to close for opportunities with valid dates. Open opportunities are not assigned artificial closing dates. The average sales cycle across analyzed closed opportunities is about 47.99 days.

9. Data Model

The Power BI report uses a relational model centered on the `sales_pipeline` fact table, with lookup tables for products, accounts, sales agents, and dates.

Table	Role	Purpose
sales_pipeline	Fact	Opportunity-level transactions and outcomes
products	Dimension	Product attributes
accounts	Dimension	Customer and industry attributes
sales_teams	Dimension	Sales agent, manager, and region
DateTable	Date dimension	Time-based reporting and DAX calculations

9.1 Relationships

Lookup column	Fact column	Type
products[product]	sales_pipeline[product]	One-to-many
accounts[account]	sales_pipeline[account]	One-to-many
sales_teams[sales_agent]	sales_pipeline[sales_agent]	One-to-many
DateTable[Date]	sales_pipeline[close_date]	Active
DateTable[Date]	sales_pipeline[engage_date]	Inactive

The active close-date relationship drives revenue trends and completed-deal analysis. The inactive engagement-date relationship supports engagement-based analysis through DAX when needed.

9.2 Modeling Decisions

- Maintain opportunity-level granularity.
- Reduce duplication of descriptive attributes.
- Support filtering by region, agent, manager, customer, and product.
- Provide consistent measures across dashboard pages.
- Enable time intelligence with a dedicated date table.

10. KPI Definitions

KPI	Definition	Result
Total Opportunities	Count of all sales opportunities	8,800
Won Opportunities	Count of opportunities in Won stage	4,238
Lost Opportunities	Count of opportunities in Lost stage	2,473
Closed Opportunities	Won + Lost	6,711
Open Opportunities	Engaging + Prospecting	2,089
Won Revenue	Sum of close values for Won opportunities	$10,005,534
Win Rate	Won / Closed opportunities	63.15%
Average Won Deal Value	Won revenue / Won opportunities	$2,360.91
Average Sales Cycle	Average engagement-to-close duration for analyzed closed deals	47.99 days
Open Pipeline Share	Open / Total opportunities	23.74%
Missing Account Share of Open Pipeline	Open opportunities missing account / Open opportunities	68.21%

11. Analysis Workflows

11.1 PostgreSQL

PostgreSQL served as a separate analytical environment to query CRM records and validate sales metrics. The analysis covered:

1. Opportunity counts and deal-stage distribution.
2. Won revenue and average deal value.
3. Closed-deal win rates.
4. Regional and product revenue.
5. Agent and manager performance.
6. Open pipeline analysis.

7. Customer and sector revenue contributions.
8. Missing account information and data quality checks.

Techniques used: relational queries, conditional aggregation, filtering, grouping, joins, and calculated metrics. Scripts are in the `PostgreSQL` folder.

11.2 Python Exploratory Data Analysis

Python independently explored the datasets, evaluated distributions, identified performance patterns, and verified key metrics.

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook in VS Code.

Workflow

1. Load the four Excel datasets.
2. Inspect dimensions and data types.
3. Check missing values and opportunity classifications.
4. Analyze opportunity distributions.
5. Compare revenue across products and regions.
6. Explore sales team and customer performance.
7. Investigate sales-cycle duration.
8. Validate results against SQL and Power BI.

11.3 Cross-Tool Validation

Results from PostgreSQL, Python, and Power BI were compared to confirm they follow the same business definitions. Validation covered total opportunity counts, won/lost/open distributions, won revenue, win rate, average won deal value, open pipeline counts, and missing account information.

The principal KPIs were consistent across all three workflows, which reduces the risk of presenting conflicting figures to stakeholders.

12. Power BI Dashboard Guide

The dashboard converts CRM opportunity data into an interactive reporting solution. It has four business-facing pages, plus a custom tooltip page and a hidden validation page.

12.1 Executive Overview

Purpose: consolidated view of overall business performance for sales leadership.

- KPI cards for won revenue, won opportunities, win rate, and other measures.
- Monthly revenue trend and month-over-month analysis.
- Product revenue comparisons.
- Regional sales performance.
- Interactive filtering and reset controls.

12.2 Sales Performance

Purpose: evaluate productivity and effectiveness of agents and managers.

- Top 10 sales agents by won revenue.
- Manager-level performance comparisons.
- Sales-cycle and win-rate comparisons.
- Revenue and opportunity visualizations.
- Custom sales agent tooltip.
- Drill-through to individual agent details.

12.3 Pipeline and Customer Analysis

Purpose: evaluate pipeline composition and customer-level performance.

- Open opportunity KPIs.
- Opportunity stages by product and region.
- Sector-level and customer account revenue.
- Missing account information analysis.
- Interactive filtering and reset controls.

12.4 Sales Agent Detail

Purpose: detailed investigation of an individual agent's performance, reached by drill-through.

- Dynamic agent selection and agent-specific KPIs.
- Monthly won revenue and product-level revenue.
- Won versus lost sales-cycle comparison.

- Opportunity-level transaction details.
- Back navigation to the originating page.

12.5 Interactive Features

Slicers and cross-filtering, custom report-page tooltips, sales agent drill-through, dynamic DAX measures, time-based analysis, reset controls, and context-sensitive visual interactions. Users can explore performance without writing SQL or Python.

13. Key Findings and Recommendations

13.1 Overall Sales Performance

Finding: $10,005,534 in won revenue from 4,238 won opportunities, with a 63.15% closed-deal win rate. The 2,089 open opportunities are 23.74% of all records.

Recommendation: review open opportunities regularly, investigate lost-deal patterns, and prioritize follow-up by value, stage, and age.

13.2 Regional Performance

Finding: West generated about $3.57M in won revenue, followed by Central at $3.35M and East at $3.09M. Central has 908 open opportunities and holds all 500 Prospecting-stage opportunities.

Recommendation: investigate Central's pipeline processes and compare stage definitions and qualification practices across regions before drawing performance conclusions.

13.3 Product Performance

Finding: GTX Pro generated about $3.51M, GTX Plus Pro $2.63M, and MG Advanced $2.22M, together about 83.6% of won revenue. GTX Basic and MG Special have 430 and 428 open opportunities respectively.

Recommendation: monitor product concentration and evaluate open opportunities by conversion rate, deal value, and pipeline age rather than counts alone.

13.4 Sales Team Performance

Finding: Darcel Schlecht generated about $1.15M in won revenue across 349 won opportunities, with a 63.11% win rate. Melvin Marxen led managers with about $2.25M in team won revenue.

Recommendation: compare agents and teams on win rate, average deal value, sales-cycle duration, opportunity volume, and revenue together, since revenue alone does not explain effectiveness.

13.5 Customer Data Quality

Finding: 1,425 of 2,089 open opportunities (68.21%) lack account information. All 1,425 are in open pipeline stages, and they represent about 16.19% of all opportunities.

Recommendation: introduce stage-appropriate account completeness checks and investigate records that progress without account identification. Missing accounts should not automatically be treated as invalid, since account identification may not be required at every sales stage.

14. Limitations

- **Historical dataset:** the data is static, not a live CRM, so results should not be read as current performance.
- **No real-time refresh:** the report uses Excel sources with no automated CRM integration or scheduled refresh.
- **Missing account information:** limits customer-level analysis of unresolved opportunities.
- **No predictive forecasting:** the project is descriptive and diagnostic only.
- **Limited business context:** no lead qualification history, marketing attribution, sales targets, or reasons for lost deals. Recommendations are analytical suggestions, not proven causal conclusions.

15. Future Improvements

1. Integrate a live CRM data source.
2. Automate data refresh and quality checks.
3. Add sales targets and quota attainment analysis.

4. Track pipeline aging and stage transitions over time.
5. Introduce forecasting and conversion probability models.
6. Expand customer segmentation and account-level analysis.
7. Implement standardized CRM data quality monitoring.

16. Reproducing the Analysis

1. Download or clone the repository and keep the folder structure intact.
2. Install the Python libraries listed in `Python/requirements.txt`.
3. Open `Python/CRM_Sales_Pipeline_EDA.ipynb` with the working directory set correctly. The notebook reads the Excel files from `Raw_Data/` using relative paths, so no drive-specific paths need editing.
4. Run the SQL scripts in the `PostgreSQL` folder to reproduce and validate the KPIs.
5. Open the Power BI report, which references the Excel files in `Power_Query/`.

17. Conclusion

The project demonstrates a complete analytical workflow: understanding business requirements, preparing raw data, validating metrics across tools, building an interactive dashboard, and communicating findings. It surfaced clear patterns in revenue, regional performance, product concentration, sales team productivity, open pipeline composition, and CRM data quality, using Excel, Power Query, PostgreSQL, Python, DAX, and Power BI.